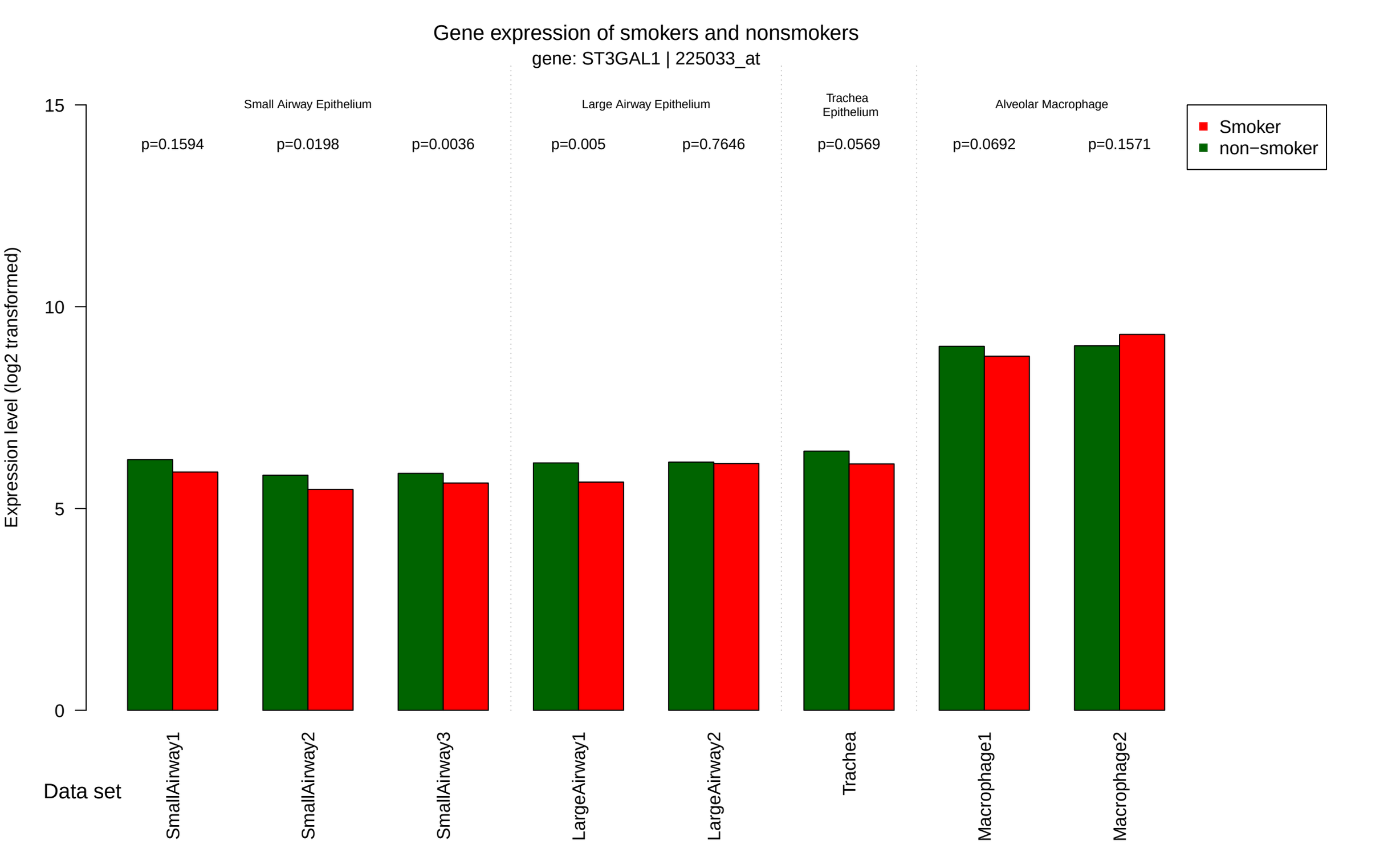
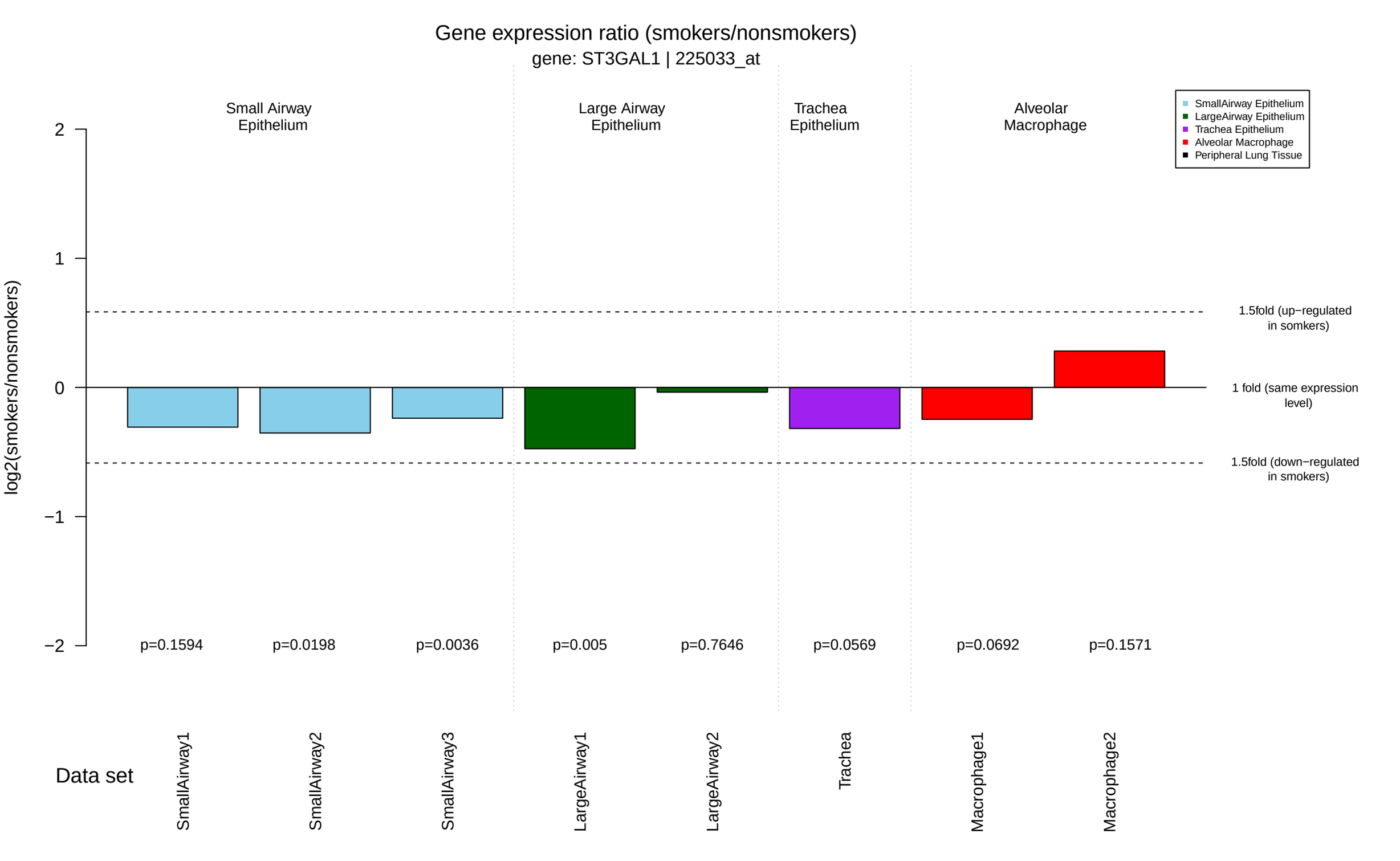
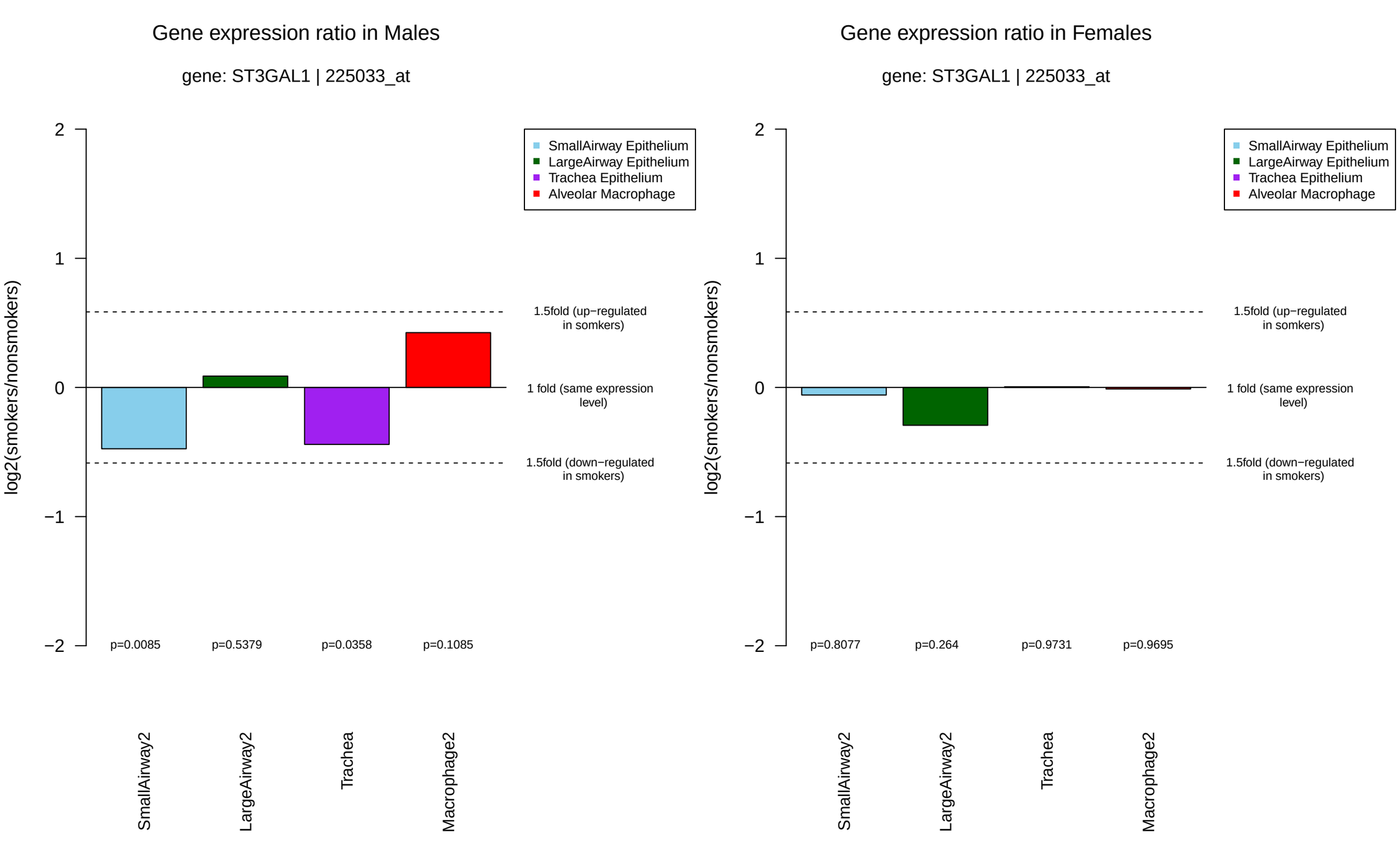
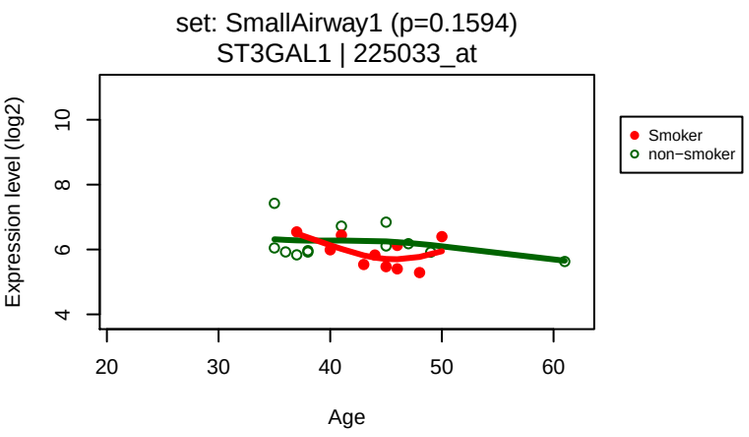
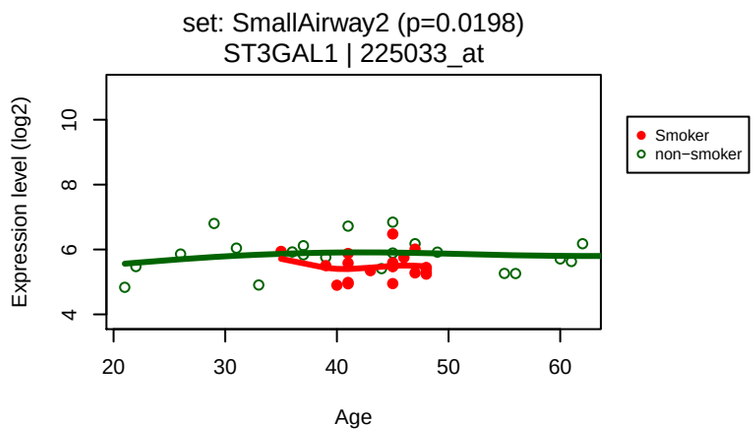
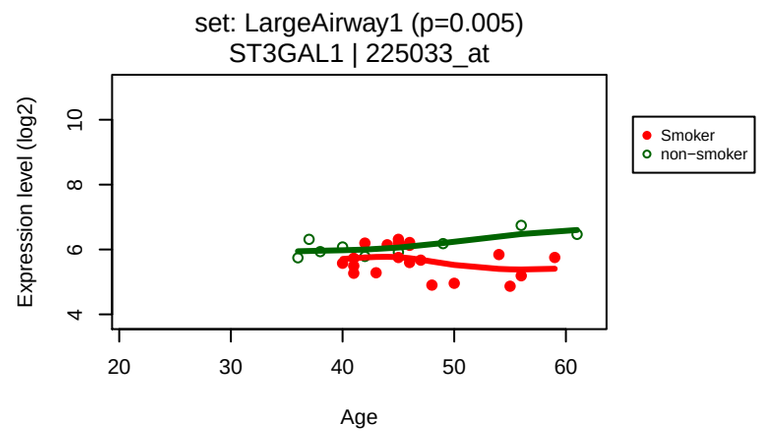

Gene expression of smokers and nonsmokers
gene: ST3GAL1 | 225033_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1594
p=0.0198
p=0.0036
p=0.005
p=0.7646
p=0.0569
p=0.0692
p=0.1571
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

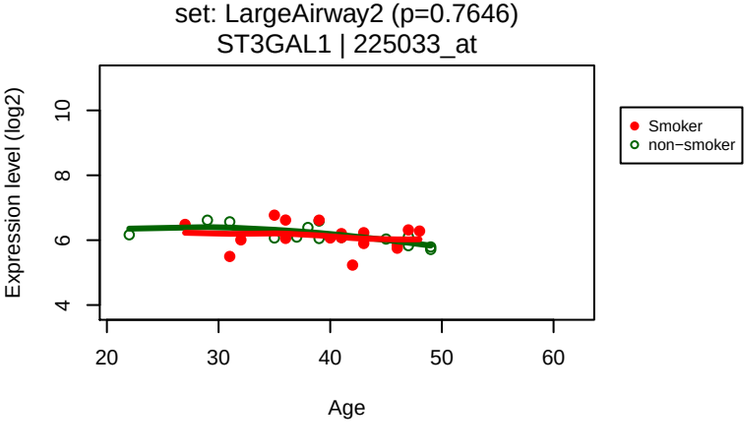

Gene expression ratio (smokers/nonsmokers)
gene: ST3GAL1 | 225033_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1594
p=0.0198
p=0.0036
p=0.005
p=0.7646
p=0.0569
p=0.0692
p=0.1571
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

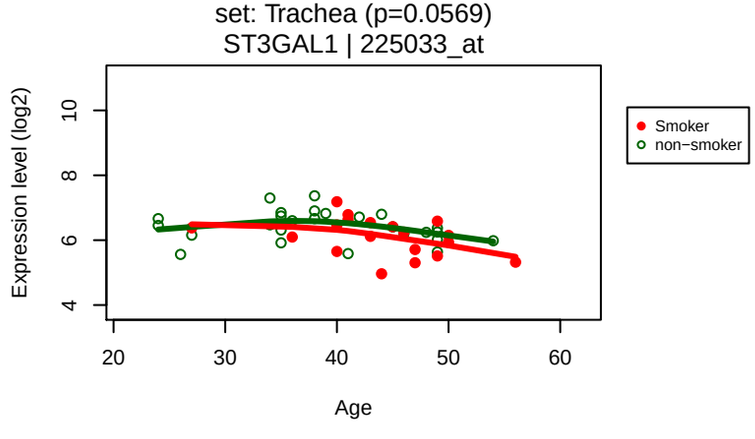

Gene expression ratio in Males
gene: ST3GAL1 | 225033_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0085
p=0.5379
p=0.0358
p=0.1085
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST3GAL1 | 225033_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8077
p=0.264
p=0.9731
p=0.9695
SmallAirway2
LargeAirway2
Trachea
Macrophage2

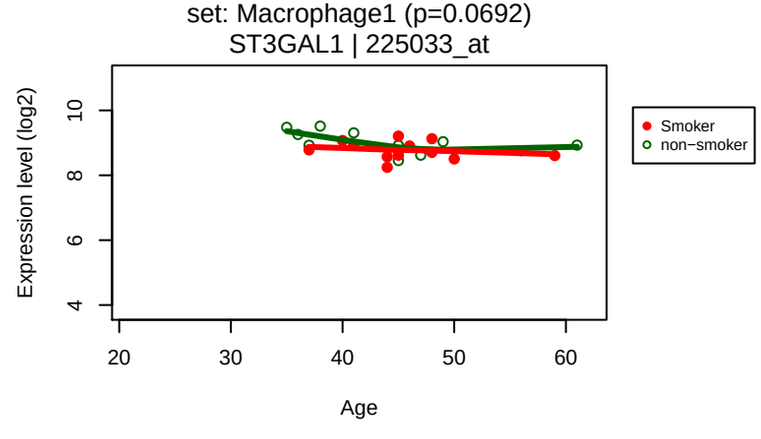

set: SmallAirway1 (p=0.1594)
ST3GAL1 | 225033_at
Expression level (log2)
Age
Smoker
non−smoker

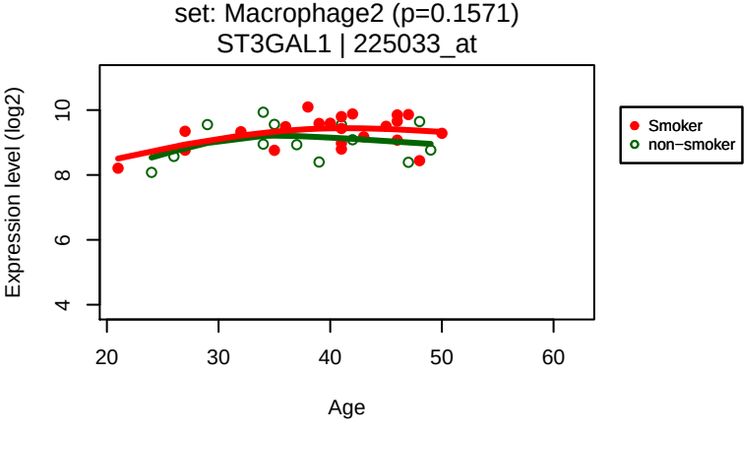

set: SmallAirway2 (p=0.0198)
ST3GAL1 | 225033_at
Expression level (log2)
Age
Smoker
non−smoker

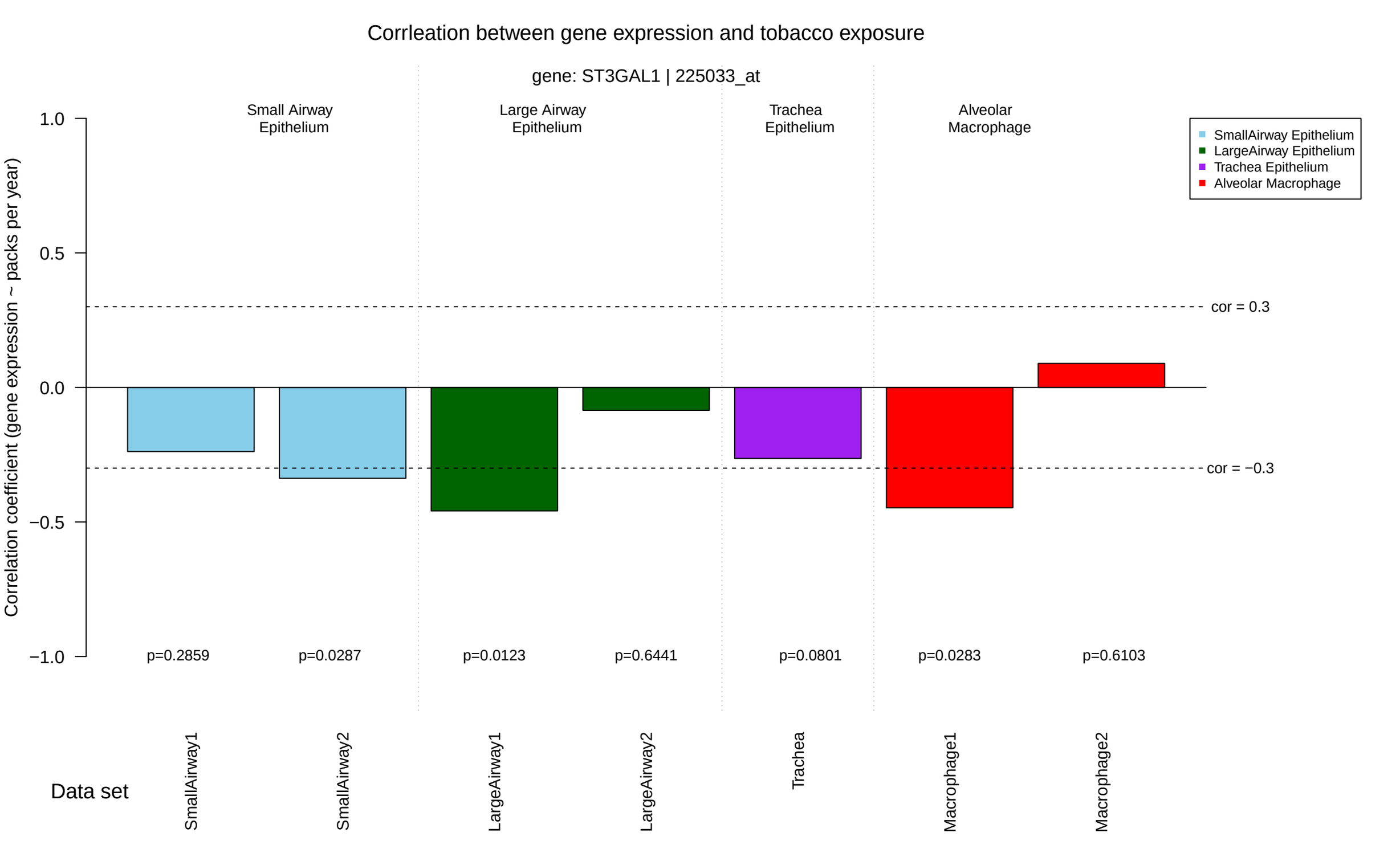

set: LargeAirway1 (p=0.005)
ST3GAL1 | 225033_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7646)
ST3GAL1 | 225033_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0569)
ST3GAL1 | 225033_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0692)
ST3GAL1 | 225033_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1571)
ST3GAL1 | 225033_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST3GAL1 | 225033_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2859
p=0.0287
p=0.0123
p=0.6441
p=0.0801
p=0.0283
p=0.6103
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2